Untitled

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0.0.1 To find the sum of series G.P

[1]:
```
print("Program To Find Sum Of G.P Series ".center(80,'='))
n = int(input("enter number of terms :- "))
x = float(input("give common ratio :- "))
s = 1*(1-(x)**(n))/(1-x)
print("Sum of given G.P is ",round(s,2))
```

```
======================Program To Find Sum Of G.P Series ======================

enter number of terms :-  5
give common ratio :-  4

Sum of given G.P is  341.0
```

0.0.2 To find the sum of 1+ 1/2! + 1/3! + 1/4! + ...1/n!

[2]:
```
print("Program to find the sum of 1+ 1/2! + 1/3! + 1/4! + ...1/n!".
 ↪center(80,'='))
n = int(input("Enter the value of n :- "))
s = 0.0
fact = 1
for i in range(1, n + 1):
    fact *= i
    s += 1 / fact
print("The sum of the series",s)
```

```
===========Program to find the sum of 1+ 1/2! + 1/3! + 1/4! + ...1/n!===========

Enter the value of n :-  6

The sum of the series 1.7180555555555557

Enter the value of n :-  10

The sum of the series 1.72
```

0.0.3 To Print Fibonacci Series [0,1,1,2,3,5,8,13]

[2]:
```
print("Program To Print Fibonacci Series [0,1,1,2,3,5,8,13] ".center(80,'='))
n = int(input("Enter the number of terms :- "))
a = 0
b = 1
print(a, end=", ")
if n > 1:
    print(b, end=", ")
for i in range(2, n):
    c = a + b
    print(c, end=", ")
    a = b
    b = c
```

```
============Program To Print Fibonacci Series [0,1,1,2,3,5,8,13] =============

Enter the number of terms :-  13

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144,

Enter the number of terms :-  8

0, 1, 1, 2, 3, 5, 8, 13,
```

0.0.4 To Check String is Palidrom or not

[3]:
```
print("Program to check string is palidrom or not".center(80,'='))
a=input('enter a string:-')
for i in range(len(a)//2):
    if a[i]!=a[-i-1]:
        print('print string is not palindrome')
        break
else:
    print('string is palindrome')
```

```
==================Program to check string is palidrom or not==================

enter a string:- (((())))

print string is not palindrome

enter a string:- naman

string is palindrome
```

0.0.5 To reverse the number

[4]:
```
print("Program to reverse number".center(80,'='))
num = int(input("Enter a number :- "))
reverse = 0
while num > 0:
```

```
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10
print("The reverse of the number is:", reverse)
```

```
===========================Program to reverse number============================

Enter a number :-  100

The reverse of the number is: 1

Enter a number :-  46

The reverse of the number is: 64
```

0.0.6 To find whether a no. is Armstrong or not

[5]:
```
print("Program to find whether a no. is Armstrong or not".center(80,'='))
n = int(input("Enter a number :- "))
s = 0
t = n
l = len(str(n))
while t > 0:
    d = t % 10
    s += d ** l
    t //= 10
if s == n:
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

```
===============Program to find whether a no. is Armstrong or not================

Enter a number :-  153

Armstrong Number

Enter a number :-  153

Armstrong Number
```

0.0.7 To count no. of vowels, no. of consonants, and no. of special symbols in a string

[6]:
```
print("Program to count no. of vowels, no. of consonants, and no. of special␣
 ↪symbols in a string".center(80,'='))
s = input("Enter a string :- ")
v = "aeiouAEIOU"
vc = 0
cc = 0
sc = 0
for i in s:
    if i.isalpha():
```

```
        if i in v:
            vc += 1
        else:
            cc += 1
    elif i.isnumeric():
        pass
    else:
        sc += 1

print("Number of vowels:", vc)
print("Number of consonants:", cc)
print("Number of special symbols:", sc)
```

```
Program to count no. of vowels, no. of consonants, and no. of special symbols in
a string

Enter a string :-  my name is lakhan

Number of vowels: 5
Number of consonants: 9
Number of special symbols: 3

Enter a string :-  Computer science is fun

Number of vowels: 8
Number of consonants: 12
Number of special symbols: 3
```

0.0.8 To input any string and count no. of words starting with a vowel

[7]:
```
print("Program to count number of words starting with a vowel. ".center(80,'='))
s = input("Enter a string :- ")
v = "aeiouAEIOU"
w = 0
words = s.split(" ")
for i in words:
    if i[0] in v:
        w += 1
print("Number of words starting with a vowel:", w)
```

```
============Program to count number of words starting with a vowel. ============

Enter a string :-  i eat apple but not mango

Number of words starting with a vowel: 3

Enter a string :-  Orange apple both are good fruit

Number of words starting with a vowel: 3
```

0.0.9 To convert upper-case to lower-case and vice-versa

[8]:
```
print("Program to convert upper-case to lower-case and vice-versa".
↪center(80,'='))
d={'a': 'A', 'A': 'a', 'b': 'B', 'B': 'b', 'c': 'C', 'C': 'c', 'd': 'D', 'D':␣
↪'d',
   'e': 'E', 'E': 'e', 'f': 'F', 'F': 'f', 'g': 'G', 'G': 'g', 'h': 'H', 'H':␣
↪'h',
   'i': 'I', 'I': 'i', 'j': 'J', 'J': 'j', 'k': 'K', 'K': 'k', 'l': 'L', 'L':␣
↪'l',
   'm': 'M', 'M': 'm', 'n': 'N', 'N': 'n', 'o': 'O', 'O': 'o', 'p': 'P', 'P':␣
↪'p',
   'q': 'Q', 'Q': 'q', 'r': 'R', 'R': 'r', 's': 'S', 'S': 's', 't': 'T', 'T':␣
↪'t',
   'u': 'U', 'U': 'u', 'v': 'V', 'V': 'v', 'w': 'W', 'W': 'w', 'x': 'X', 'X':␣
↪'x',
   'y': 'Y', 'Y': 'y', 'z': 'Z', 'Z': 'z'}
s = input("Enter a string :- ")
out = ""
for i in s:
    if i in d:
        out += d[i]
    else:
        out += i
print("Converted string:", out)
```

```
===========Program to convert upper-case to lower-case and vice-versa===========

Enter a string :-  KHul Ja SiM SIm

Converted string: khUL jA sIm siM

Enter a string :-  I am CLASS 11th STUDENT

Converted string: i AM class 11TH student
```

0.0.10 To print the pattern

[9]:
```
print("Program to print parttern".center(80,'='))
n = int(input("Enter length of parttern :- "))
l = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
for i in range(n):
    for j in range(n-i):
        print(l[j], end="")
    print()
```

```
===========================Program to print parttern============================

Enter length of parttern :-  9
```

```
ABCDEFGHI
ABCDEFGH
ABCDEFG
ABCDEF
ABCDE
ABCD
ABC
AB
A
```

```
Enter length of parttern :-  5
```

```
ABCDE
ABCD
ABC
AB
A
```

0.0.11 To print the pattern

[10]:
```python
print("Program to print parttern".center(80,'='))
n = int(input("Enter length of parttern :- "))
l = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
for i in range(n):
    print(l[0:i])
```

```
===========================Program to print parttern============================
```

```
Enter length of parttern :-  7
```

```

A
AB
ABC
ABCD
ABCDE
ABCDEF


A
AB
ABC
ABCD
```

0.0.12 To print the pattern

[11]:
```python
print("Program to print parttern".center(80,'='))
n = int(input("Enter length of parttern :- "))
l = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
for i in range(n):
```

```
    print(l[n-i-1]*(n-i))
```

```
==========================Program to print parttern===========================

Enter length of parttern :-  5

EEEEE
DDDD
CCC
BB
A

EEEEE
DDDD
CCC
BB
A
```

0.0.13 To print the pattern

[12]:
```
print("Program to print parttern".center(80,'='))
n = int(input("Enter length of parttern :- "))
for i in range(n):
    print("*"*i,end="")
    print()
```

```
==========================Program to print parttern===========================

Enter length of parttern :-  12


*
**
***
****
*****
******
*******
********
*********
**********
***********


*
**
***
****
```

0.0.14 To find minimum & maximum value from a list without using function

[13]:
```
print("Program to find minimum & maximum value from a list without using␣
↪function".center(80,'='))
n = int(input("how many numbers do you want to enter :- "))
numbers = []
for i in range(n):
    x = int(input("enter one number of list"))
    numbers.append(x)
mini = numbers[0]
maximum = numbers[0]
for i in numbers:
    if i < mini:
        mini = i
    if i > maximum:
        maximum = i
print("Minimum:", mini)
print("Maximum:", maximum)
```

```
===Program to find minimum & maximum value from a list without using function===

how many numbers do you want to enter :-  5
enter one number of list 1
enter one number of list 4
enter one number of list 9
enter one number of list 15
enter one number of list 6

Minimum: 1
Maximum: 15

Minimum: 1
Maximum: 9
```

0.0.15 To find minimum & maximum value from a list using function

[14]:
```
print(" Program to find minimum & maximum value from a list using function".
↪center(80,'='))
n = int(input("how many numbers do you want to enter :- "))
numbers = []
for i in range(n):
    x = int(input("enter one number of list"))
    numbers.append(x)
mini = min(numbers)
maximum = max(numbers)
print("Minimum:", mini)
print("Maximum:", maximum)
```

```
====== Program to find minimum & maximum value from a list using function=======
```

```
how many numbers do you want to enter :-  5
enter one number of list 4
enter one number of list 15
enter one number of list 1
enter one number of list 9
enter one number of list 6
```

```
Minimum: 1
Maximum: 15
```

```
Minimum: 1
Maximum: 9
```

0.0.16 To find a particular value in a list. If the value is found, it should show the message "Value found at 'x' index", else show "Not found

[15]:
```
print("Program to find a particular value in a list".center(80,'='))
n = int(input("how many numbers do you want to enter :- "))
numbers = []
for i in range(n):
    x = int(input("enter one number of list :- "))
    numbers.append(x)
num = int(input("enter number to find in list :- "))
for i in range(len(numbers)):
    if numbers[i] == num:
        print("Value found at", i, "index")
        break
else:
    print("Not found")
```

```
==================Program to find a particular value in a list==================
```

```
how many numbers do you want to enter :-  7
enter one number of list :-  12
enter one number of list :-  6
enter one number of list :-  10
enter one number of list :-  16
enter one number of list :-  9
enter one number of list :-  21
enter one number of list :-  4
enter number to find in list :-  9
```

```
Value found at 4 index
```

```
Value found at 2 index
```

0.0.17 To count the frequency of elements in a list

[16]:
```
print("Program to count the frequency of elements in a list".center(80,'='))
n = int(input("how many numbers do you want to enter :- "))
nums = []
for i in range(n):
    x = int(input("enter one number of list :- "))
    nums.append(x)
dict = {}
for i in nums:
    if i in dict:
        dict[i] += 1
    else:
        dict[i] = 1
print("frequency of elements", dict)
```

```
==============Program to count the frequency of elements in a list==============

how many numbers do you want to enter :-  7
enter one number of list :-  4
enter one number of list :-  3
enter one number of list :-  9
enter one number of list :-  4
enter one number of list :-  18
enter one number of list :-  21
enter one number of list :-  18

frequency of elements {4: 2, 3: 1, 9: 1, 18: 2, 21: 1}

frequency of elements {1: 1, 2: 2, 3: 1, 5: 1}
```

0.0.18 To multiply odd No. with 2 and divide even No. by 2 in a list

[17]:
```
print("Program to multiply odd No. with 2 and divide even No. by 2 in a list".
 ↪center(80,'='))
n = int(input("how many numbers do you want to enter :- "))
nums = []
for i in range(n):
    x = int(input("enter one number of list :- "))
    nums.append(x)
modl = []
for i in nums:
    if i % 2 == 0:
        modl.append(i // 2)
    else:
        modl.append(i * 2)
print("Modified List:", modl)
```

```
=====Program to multiply odd No. with 2 and divide even No. by 2 in a list======
```

```
how many numbers do you want to enter :-  5
enter one number of list :-  2
enter one number of list :-  52
enter one number of list :-  17
enter one number of list :-  27
enter one number of list :-  16

Modified List: [1, 26, 34, 54, 8]

Modified List: [2, 1, 6, 90, 38, 17, 6, 30]
```

0.0.19 To shift all the elements to left by one in a list.

[18]:
```python
print("Program to shift all the elements to left by one in a list".
 ↪center(80,'='))
n = int(input("how many numbers do you want to enter :- "))
nums = []
for i in range(n):
    x = int(input("enter one number of list :- "))
    nums.append(x)
fe = nums.pop(0)
nums.append(fe)
print("Shifted List:", nums)
```

```
===========Program to shift all the elements to left by one in a list===========

how many numbers do you want to enter :-  6
enter one number of list :-  2
enter one number of list :-  4
enter one number of list :-  5
enter one number of list :-  3
enter one number of list :-  7
enter one number of list :-  8

Shifted List: [4, 5, 3, 7, 8, 2]

Shifted List: [2, 3, 4, 5, 1]
```

0.0.20 To calculate mean, median and mode of this list

[19]:
```python
print("Program to calculate mean, median and mode of this list".center(80,'='))
n = int(input("how many numbers do you want to enter :- "))
nums = []
for i in range(n):
    x = int(input("enter one number of list :- "))
    nums.append(x)
# Mean
SUM = 0
for i in nums:
    SUM += i
```

```
mean = SUM / len(nums)
print("mean =", round(mean,2))
# Median
l2 = sorted(nums)
lenl = len(l2)
if lenl % 2 == 0:
    median = (l2[lenl // 2 - 1] + l2[lenl // 2]) / 2
else:
    median = l2[n // 2]
print("median =", round(median,2))
# Mode
dict = {}
for i in nums:
    if i in dict:
        dict[i] += 1
    else:
        dict[i] = 1
x = 0
for i in dict:
    if dict[i] > x:
        x = dict[i]
        mode = i
print("mode =", mode)
```

```
============Program to calculate mean, median and mode of this list=============

how many numbers do you want to enter :-  6
enter one number of list :-  12
enter one number of list :-  13
enter one number of list :-  14
enter one number of list :-  12
enter one number of list :-  11
enter one number of list :-  16

mean = 13.0
median = 12.5
mode = 12

mean = 4.0
median = 4
mode = 3
```

0.0.21 To display the names of students who have marks above 75

[20]:
```
print("Program To display the names of students who have marks above 75".
 ↪center(80,'='))
n = int(input("Enter the number of students :- "))
stu = {}
for i in range(n):
```

```
    rn = input("Enter Roll Number of Student "+ str(i+1)+" :- ")
    name = input("Enter Name of Student "+ str(i+1)+" :- ")
    marks = float(input("Enter Marks of Student "+ str(i+1)+" :- "))
    stu[rn] = {"name": name, "marks": marks}

print("\nStudents with marks above 75:")
for i in stu.values():
    if i["marks"] > 75:
        print(i["name"])
```

```
========Program To display the names of students who have marks above 75========

Enter the number of students :-  5
Enter Roll Number of Student 1 :-  1
Enter Name of Student 1 :-  nobita
Enter Marks of Student 1 :-  0
Enter Roll Number of Student 2 :-  2
Enter Name of Student 2 :-  dekisuki
Enter Marks of Student 2 :-  99.99
Enter Roll Number of Student 3 :-  3
Enter Name of Student 3 :-  doremon
Enter Marks of Student 3 :-  101
Enter Roll Number of Student 4 :-  4
Enter Name of Student 4 :-  suniyo
Enter Marks of Student 4 :-  82
Enter Roll Number of Student 5 :-  5
Enter Name of Student 5 :-  gian
Enter Marks of Student 5 :-  35


Students with marks above 75:
dekisuki
doremon
suniyo


Students with marks above 75:
Amit
Arun
```

0.0.22 State and Capital

[21]:
```
print("Program To input a state name from user and print corresponding capital".
 ↪center(80,'='))
dict ={
    "Andhra Pradesh": "Amaravati",
    "Arunachal Pradesh": "Itanagar",
    "Assam": "Dispur",
    "Bihar": "Patna",
```

```
    "Chhattisgarh": "Raipur",
    "Goa": "Panaji",
    "Gujarat": "Gandhinagar",
    "Haryana": "Chandigarh",
    "Himachal Pradesh": "Shimla",
    "Jharkhand": "Ranchi",
    "Karnataka": "Bengaluru",
    "Kerala": "Thiruvananthapuram",
    "Madhya Pradesh": "Bhopal",
    "Maharashtra": "Mumbai",
    "Manipur": "Imphal",
    "Meghalaya": "Shillong",
    "Mizoram": "Aizawl",
    "Nagaland": "Kohima",
    "Odisha": "Bhubaneswar",
    "Punjab": "Chandigarh",
    "Rajasthan": "Jaipur",
    "Sikkim": "Gangtok",
    "Tamil Nadu": "Chennai",
    "Telangana": "Hyderabad",
    "Tripura": "Agartala",
    "Uttar Pradesh": "Lucknow",
    "Uttarakhand": "Dehradun",
    "West Bengal": "Kolkata"
}

s = input("Enter the name of the state :- ")
if s in dict:
    print("The capital of ", s , "is ",dict[s])
else:
    print("State not found.")
```

```
====Program To input a state name from user and print corresponding capital=====

Enter the name of the state :-  Sikkim

The capital of  Sikkim is  Gangtok

The capital of  Bihar is  Patna
```

0.0.23 Car manufacture name to car name

[22]:
```
print("Program To input car manufacture name and print the names of car".
 ↪center(80,'='))
Vehicle = {'Santro':'Hyundai', 'Tiago':'TATA', 'Nexon':'Tata', 'Safari':'tata'}
while True:
    manu = input("Enter the car manufacturer's name :- ").strip()
    manul = manu.lower()
    found = False
```

```python
    x=1
    for car, maker in Vehicle.items():
        if maker.lower() == manul:
            if not found:
                print("cars made by this company are :")
            print(x,car)
            x+=1
            found = True

    if not found:
        print("No cars found for the given manufacturer.")
    if input("Do you want to continue (y/n):- ").strip() == 'n':
        break
```

```
========Program To input car manufacture name and print the names of car========

Enter the car manufacturer's name :-  tata

cars made by this company are :
1 Tiago
2 Nexon
3 Safari

Do you want to continue (y/n):-  y
Enter the car manufacturer's name :-  TatA

cars made by this company are :
1 Tiago
2 Nexon
3 Safari

Do you want to continue (y/n):-  n

Tiago
Nexon
Safari

Do you want to continue (y/n):-  y
Enter the car manufacturer's name :-  TATA

Tiago
Nexon
Safari

Do you want to continue (y/n):-  n
```

[23]:

```python

```